

Sampled evaluation — $t = 2015$ (5:1 negative ratio, official test set)

Method	PR-AUC	AUROC	F1	NDCG@20	Prec@20	CWR
RCA Persistence	0.520	0.651	0.436	0.501	0.479	0.336
Density	0.349	0.779	0.420	0.481	0.440	0.857
ECI	0.137	0.482	0.260	—	—	0.464
KNN (LLM embeddings)	0.249	0.646	0.319	0.349	0.324	0.661
XGBoost (unoptimized)	0.643	0.896	0.628	0.679	0.621	0.935
XGBoost (optimized)	0.663	0.902	0.637	0.701	0.641	0.943
GNN-4F	0.412	0.819	0.471	0.437	0.401	0.884
GNN-11F	0.436	0.829	0.481	0.475	0.429	0.895
GNN-11F+LLM	0.440	0.832	0.484	0.478	0.427	0.899
GNN-11F-LLM + RCA-skip (unoptimized)	0.640	0.893	0.628	0.681	0.625	0.927
GNN-11F-LLM + RCA-skip (optimized)	0.641	0.895	0.626	0.673	0.614	0.935

Bold row = best method overall (wins the most metrics: XGBoost (optimized)). ECI has no within-country ranking signal, so NDCG@20 / Prec@20 are marked — rather than reported off an arbitrary tie-break.